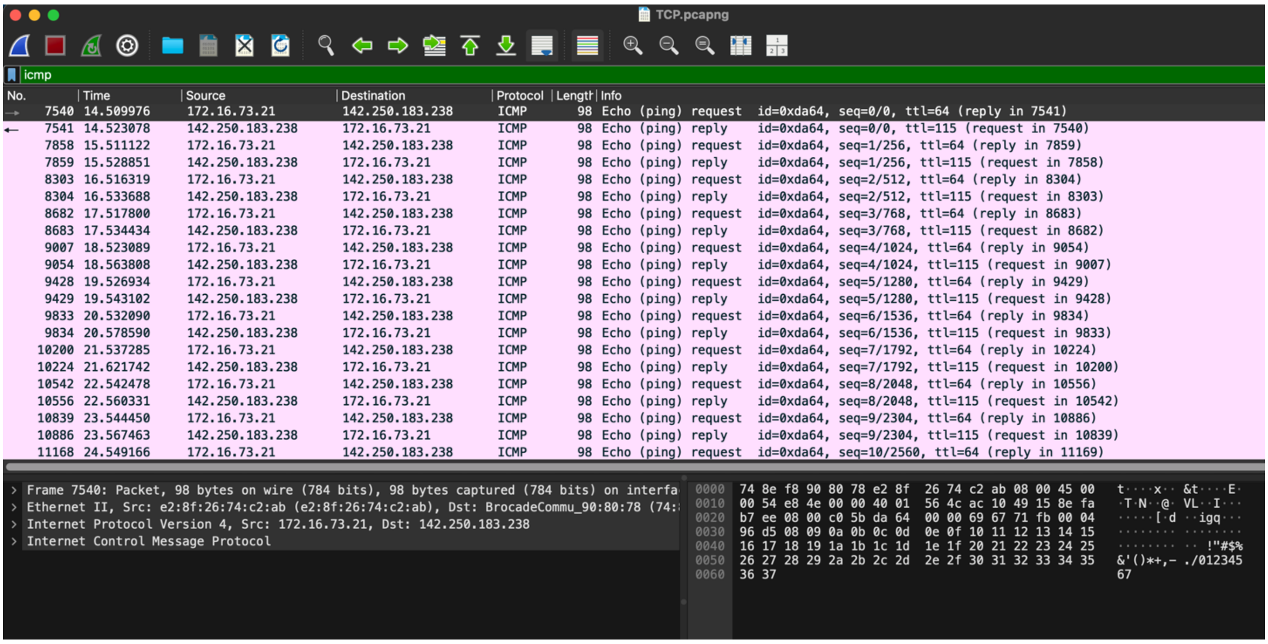
ASSIGNMENT 2

1. To capture and analyze ICMP packets (Ping request and reply)

```
harsh — -zsh — 100×35
(base) harsh@Harshs-MacBook-Air ~ % ping google.com
PING google.com (142.250.182.238): 56 data bytes
64 bytes from 142.250.182.238: icmp_seq=0 ttl=118 time=16.323 ms
64 bytes from 142.250.182.238: icmp_seq=1 ttl=118 time=59.844 ms
64 bytes from 142.250.182.238: icmp_seq=2 ttl=118 time=15.463 ms
64 bytes from 142.250.182.238: icmp_seq=3 ttl=118 time=14.400 ms
64 bytes from 142.250.182.238: icmp_seq=4 ttl=118 time=14.795 ms
64 bytes from 142.250.182.238: icmp_seq=5 ttl=118 time=51.621 ms
64 bytes from 142.250.182.238: icmp_seq=6 ttl=118 time=95.490 ms
64 bytes from 142.250.182.238: icmp_seq=7 ttl=118 time=14.350 ms
64 bytes from 142.250.182.238: icmp_seq=8 ttl=118 time=14.314 ms
64 bytes from 142.250.182.238: icmp_seq=9 ttl=118 time=16.518 ms
64 bytes from 142.250.182.238: icmp_seq=10 ttl=118 time=15.811 ms
64 bytes from 142.250.182.238: icmp_seq=11 ttl=118 time=18.380 ms
^C
--- google.com ping statistics ---
12 packets transmitted, 12 packets received, 0.0% packet loss
round-trip min/avg/max/stddev = 14.314/28.942/95.490/25.026 ms
(base) harsh@Harshs-MacBook-Air ~ %
```



TCP.pcapng

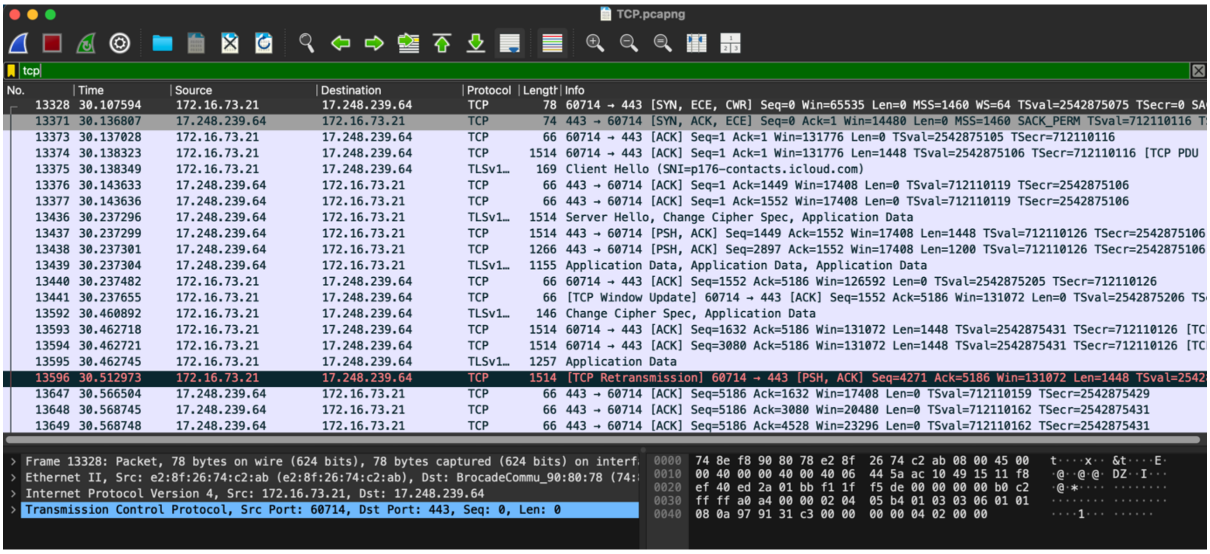
icmp

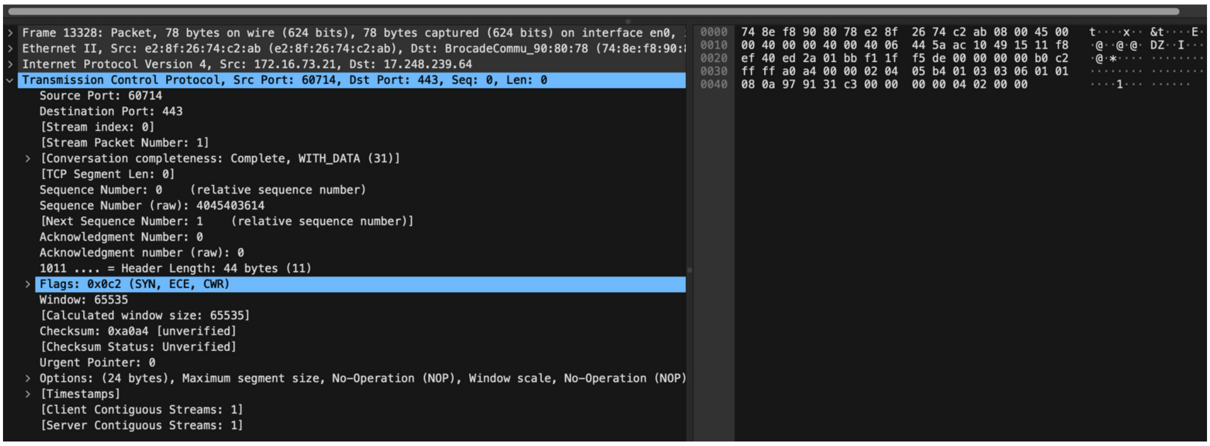
No.	Time	Source	Destination	Protocol	Length	Info
7540	14.509976	172.16.73.21	142.250.183.238	ICMP	98	Echo (ping) request id=0xda64, seq=0/0, ttl=64 (reply in 7541)
7541	14.523078	142.250.183.238	172.16.73.21	ICMP	98	Echo (ping) reply id=0xda64, seq=0/0, ttl=115 (request in 7540)
7858	15.511122	172.16.73.21	142.250.183.238	ICMP	98	Echo (ping) request id=0xda64, seq=1/256, ttl=64 (reply in 7859)
7859	15.528851	142.250.183.238	172.16.73.21	ICMP	98	Echo (ping) reply id=0xda64, seq=1/256, ttl=115 (request in 7858)
8303	16.516319	172.16.73.21	142.250.183.238	ICMP	98	Echo (ping) request id=0xda64, seq=2/512, ttl=64 (reply in 8304)
8304	16.533688	142.250.183.238	172.16.73.21	ICMP	98	Echo (ping) reply id=0xda64, seq=2/512, ttl=115 (request in 8303)
8682	17.517800	172.16.73.21	142.250.183.238	ICMP	98	Echo (ping) request id=0xda64, seq=3/768, ttl=64 (reply in 8683)
8683	17.534434	142.250.183.238	172.16.73.21	ICMP	98	Echo (ping) reply id=0xda64, seq=3/768, ttl=115 (request in 8682)
9007	18.523089	172.16.73.21	142.250.183.238	ICMP	98	Echo (ping) request id=0xda64, seq=4/1024, ttl=64 (reply in 9054)
9054	18.563808	142.250.183.238	172.16.73.21	ICMP	98	Echo (ping) reply id=0xda64, seq=4/1024, ttl=115 (request in 9007)
9428	19.526934	172.16.73.21	142.250.183.238	ICMP	98	Echo (ping) request id=0xda64, seq=5/1280, ttl=64 (reply in 9429)
9429	19.543102	142.250.183.238	172.16.73.21	ICMP	98	Echo (ping) reply id=0xda64, seq=5/1280, ttl=115 (request in 9428)
9833	20.532090	172.16.73.21	142.250.183.238	ICMP	98	Echo (ping) request id=0xda64, seq=6/1536, ttl=64 (reply in 9834)
9834	20.578590	142.250.183.238	172.16.73.21	ICMP	98	Echo (ping) reply id=0xda64, seq=6/1536, ttl=115 (request in 9833)
10200	21.537285	172.16.73.21	142.250.183.238	ICMP	98	Echo (ping) request id=0xda64, seq=7/1792, ttl=64 (reply in 10224)
10224	21.621742	142.250.183.238	172.16.73.21	ICMP	98	Echo (ping) reply id=0xda64, seq=7/1792, ttl=115 (request in 10200)
10542	22.542478	172.16.73.21	142.250.183.238	ICMP	98	Echo (ping) request id=0xda64, seq=8/2048, ttl=64 (reply in 10556)
10556	22.560331	142.250.183.238	172.16.73.21	ICMP	98	Echo (ping) reply id=0xda64, seq=8/2048, ttl=115 (request in 10542)
10839	23.544450	172.16.73.21	142.250.183.238	ICMP	98	Echo (ping) request id=0xda64, seq=9/2304, ttl=64 (reply in 10886)
10886	23.567463	142.250.183.238	172.16.73.21	ICMP	98	Echo (ping) reply id=0xda64, seq=9/2304, ttl=115 (request in 10839)
11168	24.549166	172.16.73.21	142.250.183.238	ICMP	98	Echo (ping) request id=0xda64, seq=10/2560, ttl=64 (reply in 11169)

```
> Frame 7540: Packet, 98 bytes on wire (784 bits), 98 bytes captured (784 bits) on interfa
> Ethernet II, Src: e2:8f:26:74:c2:ab (e2:8f:26:74:c2:ab), Dst: BrocadeCommu_90:80:78 (74:
> Internet Protocol Version 4, Src: 172.16.73.21, Dst: 142.250.183.238
> Internet Control Message Protocol
```

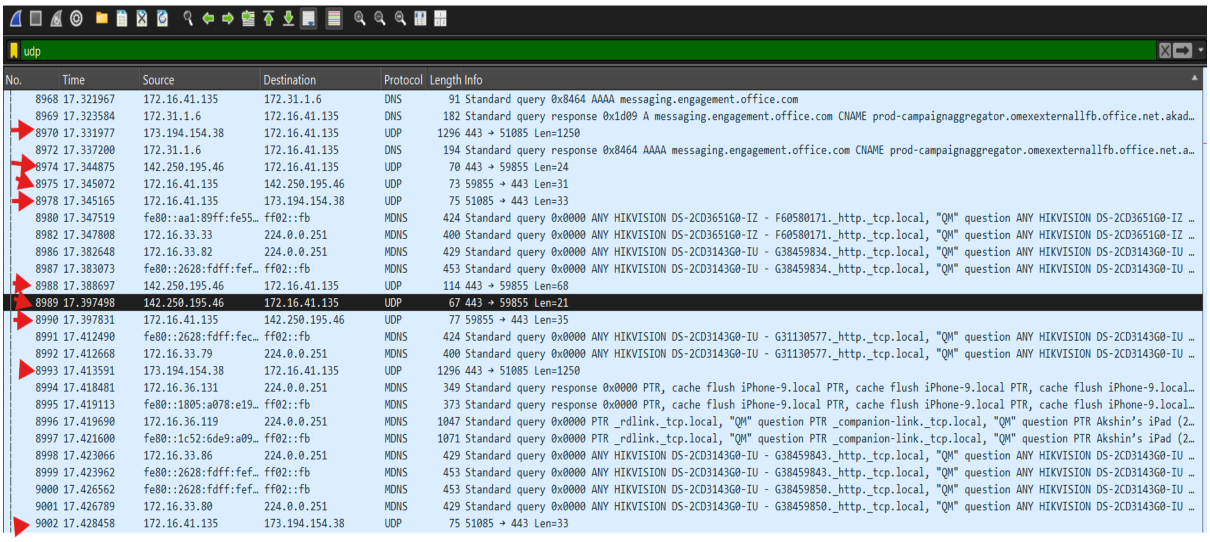
```
0000  74 8e f8 90 80 78 e2 8f  26 74 c2 ab 08 00 45 00   t····x·· &t····E·
0010  00 54 e8 4e 00 00 40 01  56 4c ac 10 49 15 8e fa   ·T·N··@· VL··I···
0020  b7 ee 08 00 c0 5b da 64  00 00 69 67 71 fb 00 04   ·····[·d ··igq···
0030  96 d5 08 09 0a 0b 0c 0d  0e 0f 10 11 12 13 14 15   ········ ········
0040  16 17 18 19 1a 1b 1c 1d  1e 1f 20 21 22 23 24 25   ········ ·· !"#$%
0050  26 27 28 29 2a 2b 2c 2d  2e 2f 30 31 32 33 34 35   &'()*+,- ./012345
0060  36 37                                              67
```

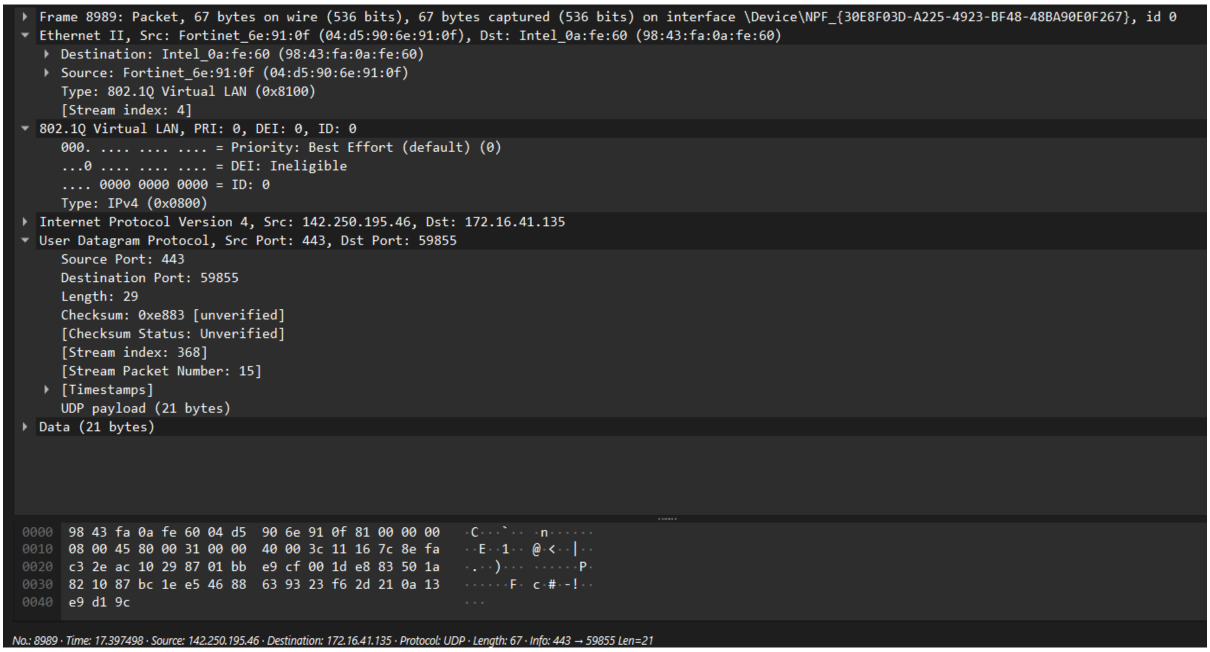
2. To analyze TCP packets and observe the TCP three-way handshake



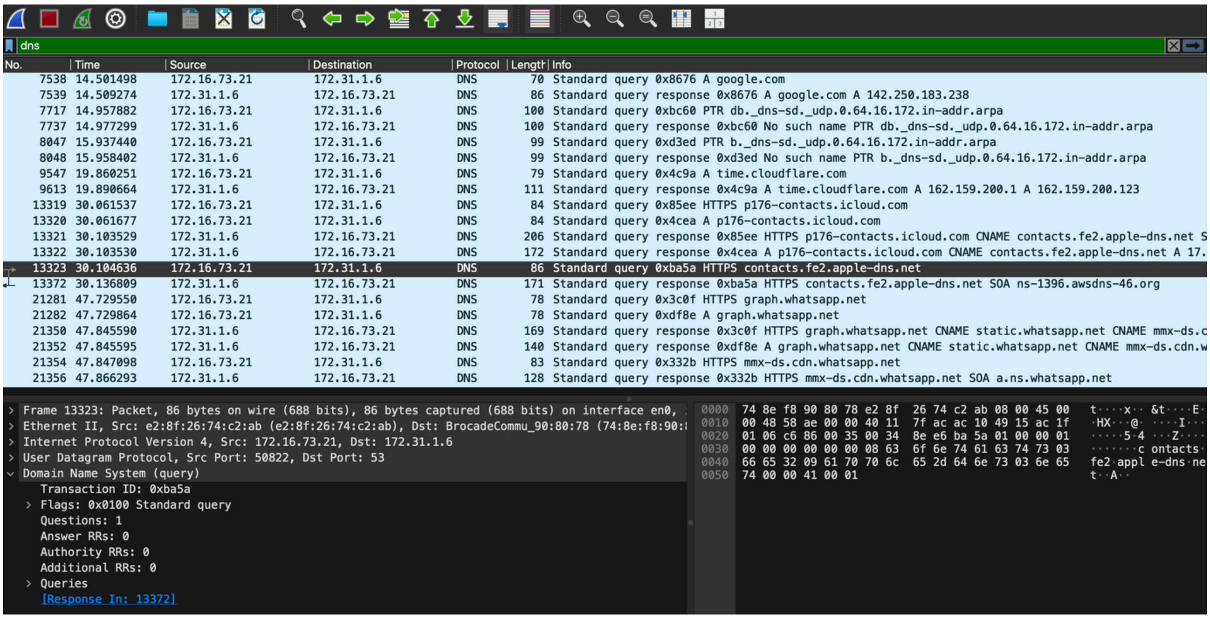


3. To capture and analyze UDP packets

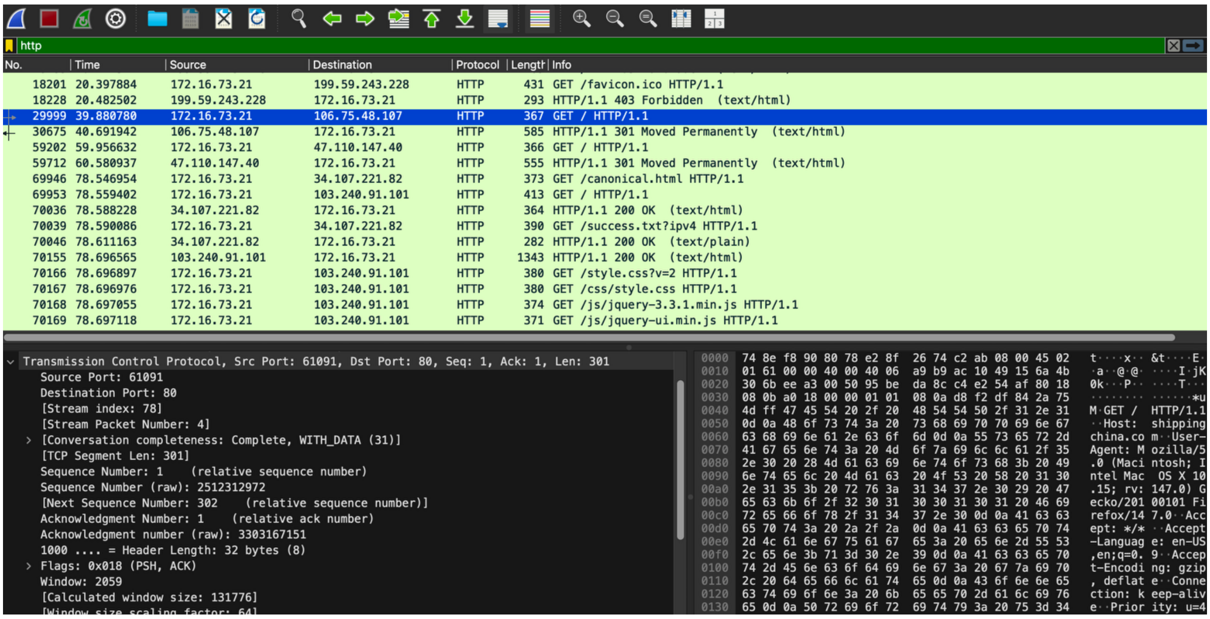




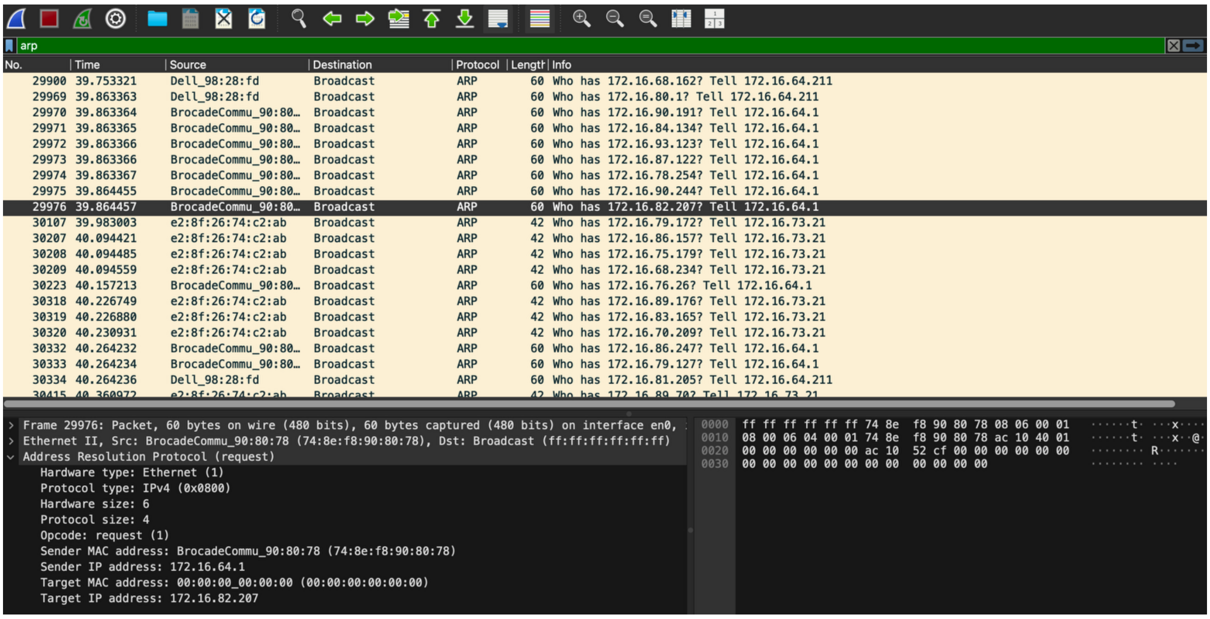
4. To capture and analyze DNS query and response packets



5. To capture and analyze HTTP packets



6. To analyze ARP request and reply packets



7. **To identify source and destination MAC and IP addresses in captured packets.**